

AI or Overhiring? Explaining the Post-Pandemic Decline in Early-Career Hiring

Literature Review

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Introduction

Recent research has raised concerns about a decline in early-career hiring, especially in industries where generative artificial intelligence can handle many tasks. Tools such as ChatGPT can assist with writing, coding, research, and information processing, which are common tasks in entry-level jobs. As these tools become more widely used, firms may need fewer new workers for certain tasks or may change the skills they look for when hiring. However, the introduction of generative AI also followed a period of unusual employment growth during the COVID-19 pandemic. Some industries that hired rapidly during the pandemic may now be reducing hiring as they return toward their earlier employment trends. This makes it difficult to determine whether the recent decline in early-career hiring is related to AI, post-pandemic employment adjustment, or both. This paper examines whether industries with greater exposure to generative AI experienced larger declines in hiring among workers ages 22 to 24 after 2022, and whether this relationship remains after accounting for employment growth above pre-COVID trends.

Literature Review

Researchers have developed several measures of workers' potential exposure to AI. Felten, Raj, and Seamans construct an occupation-level measure based on the relationship between AI applications and occupational abilities, while Eloundou et al. assess the share of tasks in each occupation that large language models could affect [1,2]. These measures show variation across occupations and industries, but they capture technological exposure rather than firms' actual AI adoption or use. High exposure therefore does not imply that jobs will necessarily disappear. If AI substitutes for labor, firms may reduce hiring for affected tasks. If it raises worker productivity or lowers production costs, firms may instead expand output, reorganize jobs, or demand different skills.

Earlier evidence shows why these boundaries matter. Using online vacancies from the period before the widespread adoption of generative AI, Acemoglu et al. find that AI-exposed establishments changed their demand for particular skills and reduced hiring in some non-AI positions, without producing a broad decline in employment [3]. Hui, Reshef, and Zhou find that freelancers in occupations more exposed to generative AI experienced reductions in employment and earnings after the release of ChatGPT [4]. At the firm and industry level, however, Liu and Webber find no evidence that measured AI adoption reduced total job postings, while emphasizing that aggregate results may conceal losses among particular occupations or groups of workers [5]. In a study of customer-support agents, Brynjolfsson, Li, and Raymond find that access to a generative AI assistant increased productivity, with larger gains among less-experienced workers [6]. Together, these findings indicate that AI's labor-market effects depend on the tasks, workers, and outcomes studied.

Workers entering the labor market may be especially exposed to hiring changes. Using the U.S. Census Bureau's Quarterly Workforce Indicators, Tucker finds an immediate and persistent decline in hiring among workers ages 22 to 24 in the industry-state cells most exposed to generative AI [7]. The subsequent employment decline operates mainly through weaker hiring rather than increased separations. Brynjolfsson, Chandar, and Chen document a related pattern in payroll data:

employment among workers ages 22 to 25 fell relative to older workers in highly exposed occupations, and the difference was concentrated in occupations where AI is more likely to substitute for human tasks [8]. They interpret these patterns as descriptive evidence rather than causal estimates. Both studies suggest that changes associated with generative AI may appear more clearly through reduced early-career hiring than through increased separations among existing workers.

The timing of these changes creates an important identification problem. The release of ChatGPT followed an unusually volatile period in which industries experienced different employment contractions, recoveries, and shifts toward remote work. Crane and Soto illustrate this problem for coding-related employment [9]. They find that total employment in coder-intensive industries is consistent with a return toward its pre-COVID trajectory after pandemic-era overhiring. At the same time, coder employment slowed relative to other occupations within the same industries, suggesting an additional occupation-specific shock. Remote-work capacity may also confound comparisons across industries. Dingel and Neiman show that the ability to work from home varies widely across occupations and industries [10], and many jobs with high generative AI exposure can also be performed remotely. Consequently, a post-2022 decline in a highly exposed industry does not by itself identify an effect of generative AI.

This paper evaluates pandemic adjustment as a competing explanation for the post-2022 relationship between AI exposure and early-career hiring. I extend Tucker's analysis by constructing an industry-level measure of how far employment before the release of ChatGPT was above or below the level predicted by its pre-COVID trend. I then examine whether the association between generative AI exposure and early-career hiring remains after accounting for this deviation. This approach does not treat AI exposure and pandemic adjustment as mutually exclusive explanations. Instead, it tests whether each remains associated with later hiring patterns when both are included in the analysis.

The analysis combines Quarterly Workforce Indicators data on workers ages 22 to 24 across industries, states, and quarters with an industry-level measure of generative AI exposure based on Eloundou et al. [2]. I will calculate industry AI exposure using occupation-level scores and pre-pandemic employment shares. I will also construct a measure of how far employment in each state-industry group was above or below its predicted pre-COVID trend. Using total employment across age groups, I will estimate expected employment from pre-COVID trends and compare it with actual employment through 2022 Q3. The analysis will then examine whether industries with greater AI exposure experienced weaker early-career hiring after accounting for this earlier employment growth. I will also compare workers ages 22 to 24 with older workers within the same industry, state, and quarter.

Skill Review

Research on AI and employment requires skills in combining different datasets and constructing measures of AI exposure. One challenge is that AI exposure is usually measured at the occupation level, while employment and hiring data may be reported by industry, state, age group, and quarter. Researchers must align occupation and industry classifications and use employment shares to calculate industry-level exposure. For example, Felten, Raj, and Seamans develop measures of AI exposure that apply across occupations and industries [1]. Eloundou et al. create occupation-level

exposure scores based on the tasks workers perform [2]. At the same time, Tucker combines AI exposure with industry and state employment data to examine changes in early-career hiring [7]. These studies show the importance of preparing data carefully and ensuring information from different sources can be compared. For my project, I will need to combine occupation-level AI exposure with industry employment shares and match the resulting scores to the Quarterly Workforce Indicators.

Another important part of this research is using statistical methods to compare employment changes over time. Panel data allow researchers to observe the same industries or states across multiple quarters. Fixed-effects models can account for differences between groups that remain relatively stable over time, while event-study methods can examine changes before and after the introduction of generative AI [7,11]. Difference-in-differences methods compare how outcomes change across groups with different exposure levels [11]. These methods are useful because employment can be affected by many factors besides AI, including economic conditions and changes in industry demand. However, they do not automatically prove that AI caused a change in hiring. Researchers must examine whether groups were already following different trends before AI's introduction and consider other possible explanations for their findings [11].

My project will also require forecasting and statistical inference skills. I plan to use pre-COVID employment data to estimate the employment level expected in 2022 if earlier trends had continued. I will compare this predicted level with actual employment to measure how far each state-industry group's employment was above or below its pre-COVID trend. A simple regression trend will be the main approach, and I may use a time-series forecasting model as an additional check [12]. I also need to understand clustered standard errors, which account for relationships between observations within the same industries or states [13]. I already have experience using Python for data cleaning, regression, visualization, and working with time-based datasets. I plan to use pandas for data cleaning and combining datasets [15], statsmodels for regression and forecasting [16], and possibly PyFixest for panel regression with fixed effects [17]. I will build on these skills by learning how to apply panel regression, fixed effects, event studies, and other statistical checks to this project.

Conclusion

Current research suggests that generative AI may be changing labor demand, but its effects differ across occupations, industries, and workers. Studies of early-career employment find weaker hiring among young workers in highly AI-exposed jobs [7,8]. However, other research shows that AI can increase worker productivity or change the skills firms demand without causing a broad decline in employment [3,5,6]. The timing of recent employment changes also makes it difficult to separate the possible effects of AI from the adjustment that followed unusually high pandemic employment [9]. My project will build on this research by examining AI exposure and pandemic employment growth together to understand the decline in early-career hiring better. Understanding these changes may help students make informed career decisions, help colleges prepare students for changes in entry-level work, and help employers understand how their hiring needs are changing. Although this project cannot establish that AI directly caused the decline in hiring, it may provide a clearer picture of how AI exposure and earlier employment growth are related to recent labor-market changes.

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